Vijana Kaushalya Maparathna

Software Engineer

✉ vijanakaush@gmail.com | 📞 +94702901355

🔗 **LinkedIn** | 💻 **Portfolio** | 🧪 **GitHub**

Flutter | Android | Kotlin | Spring Boot | Java Script

Senior Mobile Application Engineer with **5+ years of experience** designing and delivering **high-performance, production-grade mobile applications** using **Flutter, React Native, and Android (Kotlin & Java)** for **Crypto**, **telecom, fintech, and SaaS platforms**. Proven ability to build **scalable, secure, and offline-capable apps** integrated with **RESTful APIs, authentication systems, and cloud-based backends**.

Strong background in **cross-platform and native mobile development**, applying **Clean Architecture, MVVM, and BLoC** to create **maintainable and testable mobile solutions**. Experienced in **CI/CD pipelines, localization (i18n), deep linking, push notifications, and real-time data synchronization and Firebase**. Known for delivering **business-critical mobile apps**, mentoring engineers, and collaborating effectively across **product, backend, and QA teams** in Agile environments.

Technical Skills

Programming Languages

Dart, Kotlin, Java, JavaScript, TypeScript

Backend & API Integration

Spring Boot, RESTful APIs, JWT Authentication, OAuth2, API Security, Offline Sync, GraphQL

Databases & Local Storage

SQLite (mobile), PostgreSQL, MySQL, MongoDB

Cloud & DevOps

CI/CD for Mobile Apps, Docker, Git, Gradle, App Build & Release Pipelines

Testing & Quality

JUnit, Mockito, Mobile App Testing, Automation Tools, Code Reviews

AI & Modern Integrations

SpringAI, ChatGPT API, Gemini API, AI-powered mobile & backend features

Tools & Workflow

GitHub, GitLab, Bitbucket, Azure, Jira, Agile (Scrum, Kanban)

Academic Details

- **2012- 2016** **Bachelor of Engineering (Hons) in Electronic and Electrical Engineering**

  2nd Upper first Division

  University of Sunderland – UK

Projects & Open source

Portfolio as a Mobile App: **Download** | Github Repositories: **View Project**

Spring boot generative AI project: *https://github.com/geekvk/springAI_apps*

Professional Experience

Senior Software Engineer | ESOFT technologies (pvt) Ltd | 2026/02 – Present

ET Cloud Pay — Payment Platform Architecture & Expansion

Flutter, MVVM, Clean Architecture, Dio, GoRouter, WebSockets, Secure Device Pairing, Payment Gateway, CI/CD

• Architected and developed a secure Flutter-based ET Cloud Pay payment application using MVVM and Clean Architecture, designed to integrate with third-party POS applications and payment terminals for card and digital payment processing.

• Designed a feature-first Clean Architecture with Presentation, Domain, and Data layers, applying ViewModel, Repository, Use Case, dependency injection, and SOLID principles to ensure scalability, testability, and maintainability across authentication, device pairing, payments, transactions, and terminal management.

• Architected a backend broker service to securely bridge third-party POS providers with ET Cloud Pay, enabling POS applications to initiate payment transactions through a persistent **WebSocket** communication channel while abstracting payment-provider and terminal integration complexity.

• Designed the end-to-end POS-to-payment transaction flow, where a third-party POS sends payment requests containing transaction details through the broker, ET Cloud Pay validates and processes the transaction through the configured payment gateway, and the broker securely routes the final transaction status back to the originating POS terminal.

• Engineered a resilient payment transaction orchestration layer supporting transaction lifecycle management, request validation, asynchronous payment processing, success/failure callbacks, timeout handling, duplicate-request prevention, idempotency, transaction status tracking, and failure recovery.

• Implemented secure POS-to-device pairing mechanisms using cryptographically generated device identifiers, authenticated pairing flows, short-lived pairing credentials, secure token exchange, and device-to-merchant association to prevent unauthorized POS terminals from initiating payment transactions.

• Designed secure communication between POS applications, the backend broker, and ET Cloud Pay using **TLS-encrypted WebSocket and REST communication, authenticated sessions, token-based authorization, request validation, and server-side transaction verification.**

• Implemented secure local credential and session management using platform-secured storage, minimizing exposure of authentication credentials, device secrets, and payment-related configuration data on POS devices.

• Implemented robust WebSocket connection management, including authentication during connection establishment, heartbeat mechanisms, reconnection strategies, connection-state handling, request correlation, message validation, and transaction timeout recovery to support unreliable POS network environments.
• Designed transaction correlation and idempotency mechanisms to ensure payment requests are processed safely without duplicate charges when POS terminals retry requests due to network interruptions or delayed responses.
• Integrated secure RESTful APIs using Dio for authentication, merchant/device management, transaction history, payment configuration, terminal registration, and backend communication, with centralized interceptors for authentication, error handling, logging, and request configuration.
• Implemented GoRouter-based navigation and ViewModel-driven UI state management to provide predictable navigation and transaction-state handling across payment initiation, processing, success, failure, transaction history, and device-management workflows.
• Established CI/CD pipelines for automated Flutter builds, static analysis, unit/widget testing, environment-specific configuration, and deployment, improving release reliability and reducing manual deployment effort.
• Designed the platform with observability and auditability in mind by maintaining transaction states, request identifiers

ESOFT Mobile POS Platform (Offline – first) Architecture & Expansion (Flutter, BLoC, Clean Architecture, Dio, GoRouter, Drift CI/CD, Hardware Integration, QR Scanning, Bill Printing, Payment Gateway)

- Spearheaded the architecture, design, and expansion of a Flutter-based Mobile POS platform, scaling the application across sales, cart, customer onboarding, register, payment, QR scanning, and invoice printing modules, improving release velocity by 40%.
- Architected a feature-first Clean Architecture codebase using BLoC, Repository pattern, Use Cases, and dependency injection, reducing bug regression by 30% while improving testability, scalability, and maintainability.
- Translated pixel-perfect Figma designs into production-grade Flutter UI, collaborating closely with UI/UX teams to achieve 100% design consistency across Android POS hardware devices and improve cashier workflow efficiency by 25%.
- Integrated secure RESTful APIs using Dio for cart operations, customer creation, register lifecycle management, payment workflows, and transaction history, achieving 99.9% API success reliability.
- Engineered a scalable payment gateway and transaction orchestration layer, supporting nested payment methods, split payments, callback handling, and rollback recovery, reducing payment failures by 35%.
- Implemented hardware integrations for QR code scanning and Bluetooth thermal printing, enabling rapid barcode-based checkout and instant receipt generation for retail store operations.

- Established and maintained CI/CD pipelines for automated build, testing, and POS device release deployments, reducing manual deployment effort by 60% and improving release confidence.
- Optimized app performance for low-spec POS terminals using widget rebuild minimization, lazy loading, caching, and efficient BLoC state transitions, reducing render latency by 45%.
- Mentored junior mobile developers, led code reviews, and standardized reusable component architecture, increasing team productivity by 30% and improving engineering consistency.

ESOFT QR Pay Platform Architecture & Expansion (Flutter, Provider, MVVM, Dio, GoRouter, CI/CD, QR Scanning, Bill Printing, Payment Gateway)

- Spearheaded the architecture, design, and development of a Flutter-based QR Pay ecosystem, delivering two production-grade applications—a QR Payment Scanner for merchants and a QR Code Generator for customers streamlining cashless payment experiences across retail environments.
- Architected a scalable MVVM-based application structure using Provider, Repository pattern, dependency injection, and feature-first modularization, improving maintainability, code reusability, and long-term scalability.
- Developed pixel-perfect Flutter user interfaces from Figma designs, collaborating closely with UI/UX teams to deliver intuitive payment workflows and consistent user experiences across Android devices.
- Integrated secure RESTful APIs using Dio for QR payment processing, merchant onboarding, transaction validation, payment confirmation, customer management, and transaction history, achieving high API reliability and seamless backend communication.
- Engineered a robust payment gateway integration supporting QR code generation, dynamic QR payments, payment verification, callback handling, transaction reconciliation, and failure recovery, improving transaction success rates and payment reliability.
- Built a high-performance QR scanning engine utilizing the device camera for real-time QR code detection, enabling fast and accurate merchant payment acceptance with minimal latency.
- Integrated Bluetooth thermal printers to generate and print professional transaction receipts instantly after successful payments, improving operational efficiency for merchants.
- Implemented local storage services using SharedPreferences and secure local persistence mechanisms to manage authentication tokens, user preferences, offline configurations, and cached transaction data, enhancing application responsiveness and offline resilience.
- Established CI/CD pipelines for automated build, testing, and deployment workflows, reducing manual release effort and enabling faster, more reliable application delivery.
- Optimized application performance through efficient state management, widget rebuild minimization, caching strategies, and asynchronous processing, delivering smooth user experiences even on low-spec Android devices.

- Collaborated with cross-functional teams, conducted code reviews, and mentored junior developers by promoting reusable architecture patterns and Flutter best practices, improving overall code quality and development productivity.

Freelance Software Engineer (Mobile development) | KAST | 2025/12 – 2026/02

- Developed a self-custodial wallet with automated stablecoin top-ups, enabling seamless funding from Binance, MetaMask, and other external wallets. Implemented using Flutter with BLoC state management, Dio, GoRouter for navigation, and Reown for secure wallet connectivity.
- Designed and implemented UI flows for critical and optional app update screens, including reminder mechanisms over configurable day intervals, using Flutter with BLoC for state management, GoRouter for navigation, and Dio for API communication

Live app : https://play.google.com/store/apps/details?id=com.kastfinance.app&hl=en

Senior Software Engineer (Fullstack mobile development) | Mobitel (pvt) Ltd | 2023/01 – 2025/12

Self-Care Platform Modernization

Flutter, BLoC, Clean Architecture, Dio and GoRouter

- Led the **Flutter-based modernization** of Mobitel's Self-Care mobile application, serving **hundreds of thousands of users** across multiple languages.
- Built and maintained a **high-performance cross-platform mobile app** using **Flutter**, integrating **secure REST APIs** via **Dio**.
- Implemented **offline-capable user preference storage** using **SQLite**, improving reliability and usability.
- Delivered **multi-language (i18n)** support across **3 languages**, improving accessibility for a **nationwide user base**.
- Integrated **Firebase Analytics** to track **user behavior, feature usage, and conversion funnels**, enabling data-driven product improvements.
- Improved app stability and navigation by implementing **GoRouter-based routing** and **BLoC-driven state management**.

🔗 Live App: https://play.google.com/store/apps/details?id=com.mobitel.selfcare

mPay – QR-Based FinTech Mobile Application - React Native, TypeScript, Redux toolkit, Axios

- Developed a **QR-based digital payment mobile app** enabling users to **scan, pay, and manage transactions** securely.
- Built the mobile application using **React Native and Redux**, ensuring **predictable state management** and **smooth real-time UI updates**.
- Integrated **secure REST APIs** for **wallet balance, transaction history, and QR payment processing**.
- Implemented **JWT-based authentication** and **secure session handling**, ensuring **financial-grade mobile security**.
- Optimized API interaction and caching to ensure **fast payment processing and offline-tolerant UX**.
- Delivered a **clean, scalable mobile architecture** aligned with **FinTech compliance and reliability standards**.
- Handle State management in **Redux toolkit** and middle ware in **redux thunk**

Live App : https://play.google.com/store/apps/details?id=family.mobitel.mpay&hl=en

Technologies: React Native, Redux, Java Spring Boot, REST APIs, JWT, Secure QR Payments

Sales Force Automation (SFA) Mobile App

Contributed to the development of a **mission-critical Sales Order, Inventory, and Invoice Management mobile system** used by Mobitel field sales teams to **manage shops, itineraries, and real-time billing**.

Mobile Architecture & Performance

- Implemented **MVVM with Clean Architecture** in **Android (Kotlin)**, reducing bug rates by **25%** and significantly improving code maintainability.
- Designed and built a **fully offline-first mobile data layer** using **Room Database**:
    - Enabled **sales orders, invoices, and shop data** to be created and edited **without network connectivity**
    - Implemented **automatic bi-directional sync** with backend services using **Retrofit & Coroutines** when connectivity is restored
    - Applied a **timestamp-based conflict resolution strategy** to ensure **data consistency**
      Reduced data-loss incidents by 90%

Core Mobile Features

- **CameraX integration** for capturing shop images and cheques
- **Bluetooth thermal printer integration** for instant invoice printing
- **PDF invoice generation and sharing**
- **Real-time sales, shop, and itinerary tracking**

Backend Contributions (Spring Boot)

- Contributed to the development of **RESTful backend services** for:
    - **Role-based access control**
    - **Invoice creation, updates, and retrieval**

    - **Shop creation with image metadata**
    - **Sales itinerary management**
- Built and optimized **Spring Boot APIs** backed by **MySQL**, ensuring **secure and reliable mobile-to-backend communication**.

Technologies:

Android (Kotlin, MVVM, LiveData, Coroutines, Flow, StateFlow, SharedFlow, Room), Retrofit, Java, Spring Boot, MySQL, REST APIs, Bluetooth Printing, CameraX

Network Provider Data Collection App

Android application for **real-time network diagnostics during voice and VoIP calls**, used to analyze **5G / LTE signal quality, call stability, and carrier performance**.

Mobile Architecture & Data Engineering

- Built the app using **Jetpack Compose** and **MVVM**, significantly improving UI stability and reducing UI-related bugs.
- Designed a **high-volume local data pipeline** using **Room Database**, capable of storing **millions of signal records** efficiently for offline-first processing.
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Background Monitoring & Call Detection

- Implemented a **BroadcastReceiver + Background Service architecture** to:
    - Detect **call start and end events** for **cellular and VoIP**
    - Automatically trigger **network data capture in the background** when a call begins
- Engineered a **persistent background service** to continuously collect **carrier signal strength, LTE/5G state, and dBm values** even when the app is not in the foreground.

Performance & Battery Optimization

- Optimized background execution using **foreground service, WorkManager, and coroutine scheduling**, reducing **battery consumption by ~90%** compared to naive background polling.
- Implemented **intelligent sampling and lifecycle-aware data capture**, enabling **high-frequency (≈50ms) network sampling** without degrading device performance.

Cloud Sync & Reliability

- Stored all captured signal data in **Room DB** and implemented **reliable background upload** to the **cloud backend** when network connectivity becomes available via **Retrofit**.

Technologies:

Android (Kotlin, Jetpack Compose, MVVM), Room Database,Retrofit, Coroutines & Flow, BroadcastReceiver, Background & Foreground Services, WorkManager, Telephony API, Dagger-Hilt, GitHub

Associate Software Engineer | EVICIO (pvt) Ltd | 2021/ 12 – 2023 /01

Employee Management System (Web Application - ReactJS)

Web-based HR platform for employee tracking and leave management approvals.

Key Contributions

- **Frontend Architecture**

    - Developed **15+ reusable components** with ReactJS and Material UI
    - Implemented **Redux Toolkit** for global state management, reducing prop drilling by 80%
    - Achieved **90% code reuse** through atomic design principles
- **Critical Features**
    - **Leave Management Module**:
        - Integrated REST APIs for leave request/approval workflows
        - Implemented **role-based access control** (RBAC) for managers/staff
        - Added **conflict detection** to prevent overlapping leave requests
    - **Performance Dashboard**:
        - Visualized employee KPIs with interactive charts (React ChartJS)
        - Enabled CSV export for HR reporting
- **Optimizations**
    - Reduced initial page load time **from 4s to 1.2s** via:
    - API response caching

Technologies: ReactJS, Material UI, Redux, Axios, Jira, GitHub.

Travel Itinerary Planner (React Native Mobile App)

Cross-platform mobile app for intelligent trip planning

Key Achievements

- **Performance-Centric Development**
    - Integrated **GraphQL** with Apollo Client, reducing API payload size by **45%** through selective field queries
    - Implemented **code-splitting** for map-related features, decreasing initial bundle size by **30%**
- **Core Features**
    - **Smart Route Engine**:
        - Google Maps API integration with **multi-waypoint optimization**
        - Real-time traffic location visualization using Maps JavaScript SDK
    - **State Management**:
        - Hybrid architecture using **Redux** + **Context API**
        - Implemented **optimistic UI updates** for seamless offline editing
- **User Experience**
    - Built **10+ reusable components** with React Native Paper (Material Design)
    - Reduced render latency to **<100ms** through React.memo optimization

Technologies: React Native, Redux, Context API, GraphQL, Axios, Google Maps API, Jira, GitHub

Associate electronic engineer | Tengri Aero Industries (Pvt) Ltd | 2017/07 – 2020/03

Tengri Cockpit Test unit (Firmware & Hardware Development)

- Designed and developed embedded hardware and firmware systems to test **Tengri Cockpit Firmware**.
- Used **C++11** on **Yocto Linux OS** running on **NXP i.MX6 SoC** to build diagnostic and test tools.
- Integrated **GStreamer** for real-time media stream validation and cockpit display simulation.
- Created test firmware to verify CAN, UART, and I²C-based communication between avionics modules.

Technologies: C++11, GStreamer, UART, CAN, I²C

Inventory Management System

Developed an internal system to manage imported PCBs, exported devices, and track their serial numbers and statuses.

Key Contributions

- **System Architecture**
  - Designed a **real-time tracking engine** that reduced device lookup time by **70%** (previously manual Excel-based)
  - Implemented **serial number validation** with regex pattern matching, eliminating 95% of data-entry errors
- **Core Features**
  - **Status Monitoring**:
    - Automated status transitions (e.g., "In Transit" → "Warehouse Received") with email/SMS alerts
    - Batch import/export via CSV for compatibility with legacy systems
  - **Reporting Dashboard**:
    - Flagged overdue shipments using **MySQL event schedulers**
- **Business Impact**
  - Reduced inventory reconciliation time from **8 hours** → **45 minutes** weekly

Technologies: Java, MySQL.

Trainee electronic engineer | Riscor consultants (Pvt) Ltd |2015/01 – 2016/ 05

Vehicle Tracking System

IoT solution for fleet location monitoring

Key Achievements

- **Hardware Integration**
  - Developed custom **Arduino-based GPS module** with:
    - NEO-6M GPS receiver (+/- 2.5m accuracy)
    - GSM/GPRS (SIM800L) for data transmission
- **Android Application**
  - Built tracking app featuring:

        - **Google Maps SDK** with live position markers
        - **Geofencing alerts** for route deviations
        - **Offline caching** using SQLite for connectivity gaps
    - Reduced battery drain by **40%** via:
        - Adaptive location polling (1s → 30s when stationary)

Technologies: Java, XML, GPS Module, Arduino.